Data-Driven Business Strategy Challenge

*Final project for **Advanced Business Analytics** course*

This project allows you to **synthesize all elements of the PDMT framework** — *Problem, Data, Method, and Tools* — in addressing a **realistic and strategic business challenge**.

You are expected to **think as data-driven business leaders/consultants** for a digital business, not just as analysts. You must **design and justify an analytics-driven initiative** that solves a *new*, *complex*, or *cross-functional* business challenge – using the **PDMT framework** to guide your plan.

The project should demonstrate how data analytics can:

- Improve business decision-making,
- Enhance or automate existing processes,
- Create new products, services, or customer experiences, or
- Generate measurable economic and strategic value.
- Etc.

Here is the overview of the project:

1. Problem – Identify a Business Transformation Opportunity

- Pick or design a business scenario where analytics could create value **across multiple domains** (e.g., marketing + customer + finance).
- Define the business problem in strategic terms:
    - What's the opportunity or pain point?
    - Who are the stakeholders?
    - What decisions could be improved with analytics?
    - Etc.

Deliverable:

- Strategic problem statement
- PDMT map at a high level (showing problem–data–method–tool connections)
- Expected business value

2. Data – Design a Data Plan

- Students must specify what data is *needed* and *enough* — even if not all exists.
- Identify potential **data sources**, integration challenges, and privacy/ethical issues.
- They may use partial real data or public data to demonstrate feasibility.

Deliverable:

- Data architecture diagram or schema
- Description of data sources and features
- Example data extract or simulated sample

3. Method – Build a Conceptual or Partial Analytical Solution *(recommended)*

- Instead of full implementation, focus on **choosing and justifying** analytical methods appropriate to the business goal.
- They may implement one component (e.g., clustering customers, forecasting demand).
- Must explain how methods fit into the decision workflow.

Deliverable:

- Analytical design (e.g., flowchart or pseudo-pipeline)
- Prototype analysis or mock results
- Partial implementation *(recommended)*
- Evaluation plan (how to measure success)

4. Tools – Construct the Analytics Architecture

- Choose and justify tools for the proposed solution
- Emphasize scalability, collaboration, and usability for non-technical managers.

Deliverable:

- Description of tool choices
- Workflow explanation

5. Communication & Strategy Pitch

- Deliver a *consulting-style pitch deck*:
    - Problem framing
    - PDMT logic
    - Expected ROI or strategic advantage
    - Implementation roadmap

Deliverable:

- slide presentation
- several-pages executive summary
- data + codes